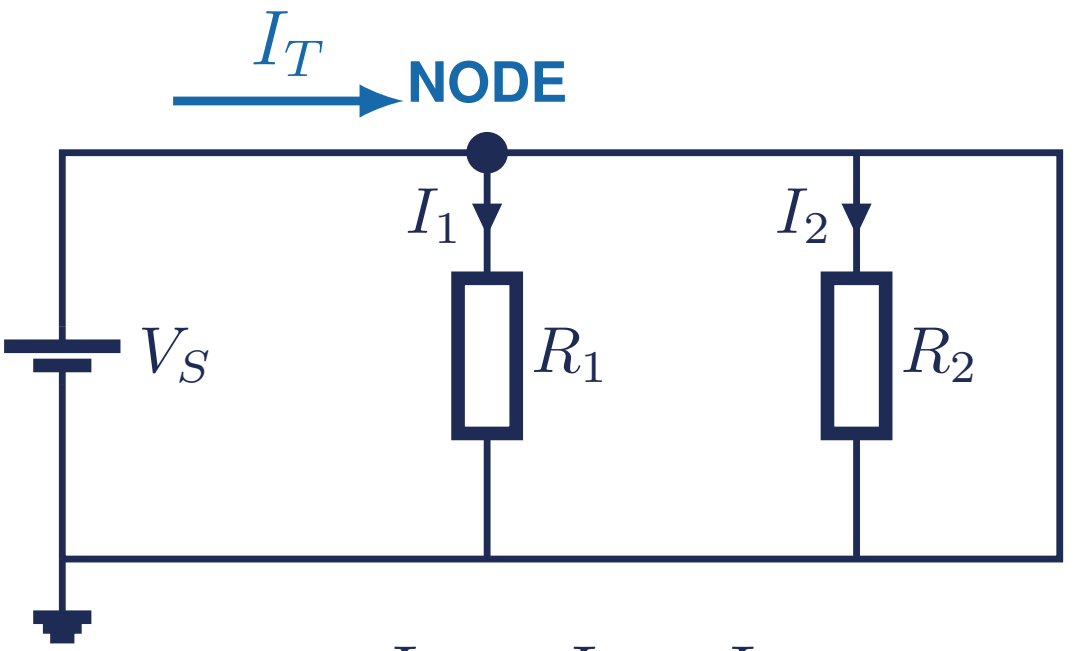
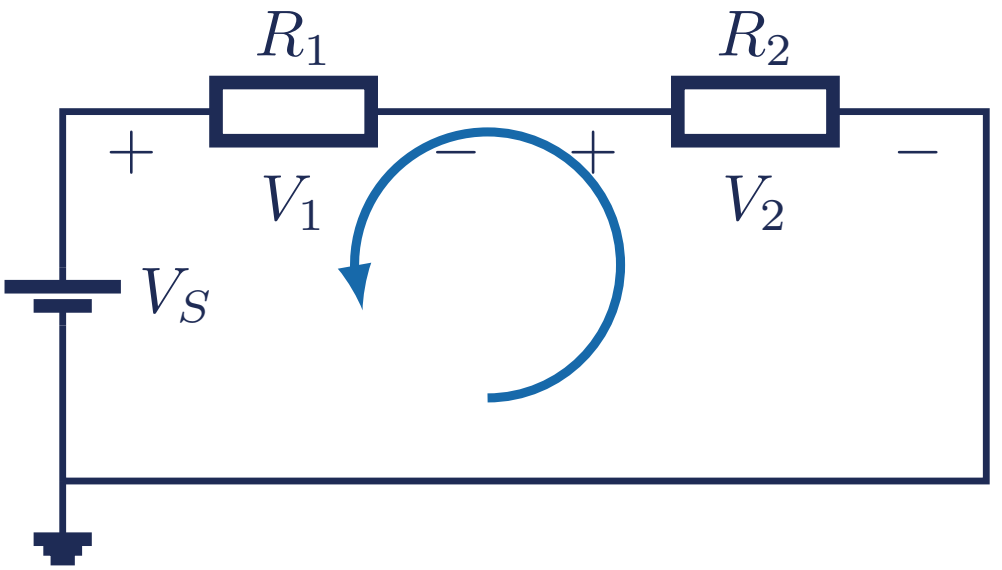
KCL at a circuit node



$$I_T = I_1 + I_2$$

$$\sum I_{\text{in}} = \sum I_{\text{out}}$$

KVL around a loop



$$V_S - V_1 - V_2 = 0 \qquad V_S = V_1 + V_2$$